

SOLUTION DESIGN DOCUMENT (SDD)

Amazon Sales Dashboard Analysis

Technology: Microsoft Power BI

Data Source: Amazon-style Sales Dataset

Data Transformation: Power Query

Analytical Language: DAX

Dashboard Pages: 3

1. Solution Overview

The Amazon Sales Dashboard is an interactive Power BI reporting solution designed to analyze sales transactions across products, categories, regions, customers, and payment methods.

The solution follows a structured data analytics workflow:

Source Data → Power Query → Data Model → Calculated Columns → DAX Measures → Dashboard Visualizations → Business Insights

The solution converts raw transactional data into a clean and interactive reporting model.

2. Technical Architecture

Data Flow

Amazon Sales Dataset

↓

Power Query

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Data Cleaning & Transformation

↓

Power BI Model

↓

Calculated Columns

↓

DAX Measures

↓

Interactive Dashboard



Business Insights

3. Source Data

The source dataset contains:

- 6,000 rows
- 10 columns
- January 2023 to December 2024 transaction data

The main fields include order details, product information, quantity, price, sales, region, customer, and payment method.

4. Power Query Transformation

Power Query is used as the data preparation layer.

Transformation Steps

4.1 Header and Data Type Validation

The first row is promoted as the table header.

Appropriate data types are assigned to:

- Order_Date → Date
- Quantity → Whole Number
- Price → Decimal Number
- Sales → Decimal Number

4.2 Text Cleaning

Unwanted spaces are removed from text fields.

Example:

" North" → "North"

"South " → "South"

4.3 Standardization

Region values are standardized so that variations such as:

north

NORTH

North

are treated consistently.

4.4 Duplicate Removal

Duplicate records are removed across the dataset.

The raw dataset contained **180 duplicate rows**.

4.5 Missing Value Handling

Missing values were identified in fields including:

- Region
- Payment_Method
- Customer_Name
- Price
- Quantity

The affected records were filtered/flagged rather than blindly assigning assumed values.

4.6 Invalid Record Filtering

Rows containing non-positive quantities are removed because they can distort revenue calculations.

4.7 Sales Recalculation

Sales is recalculated using:

Sales = Quantity × Price

This ensures consistency between quantity, price, and revenue.

5. Calculated Columns

The following calculated columns are included in the Power BI model.

Sales (Clean)

Sales = Amazon_Data[Quantity] * Amazon_Data[Price]

Purpose:

Ensures that sales are calculated consistently from quantity and price.

Order Month

Order Month = FORMAT(Amazon_Data[Order_Date], "MMM YYYY")

Purpose:

Provides a month-year field for dashboard analysis and filtering.

Order Year

Order Year = YEAR(Amazon_Data[Order_Date])

Purpose:

Supports yearly analysis.

Price Tier

Price Tier =

```
IF(
    Amazon_Data[Price] >= 3000,
    "Premium",
    IF(
        Amazon_Data[Price] >= 1000,
        "Mid",
        "Budget"
    )
)
```

Purpose:

Classifies products according to their unit price.

Region Clean

Region Clean = TRIM(PROPER(Amazon_Data[Region]))

Purpose:

Provides a standardized region field.

Is Weekend

Is Weekend =

```
IF(
    WEEKDAY(Amazon_Data[Order_Date], 2) > 5,
    "Weekend",
    "Weekday"
)
```

Purpose:

Identifies weekend and weekday orders.

6. DAX Measures

The dashboard uses reusable DAX measures for KPI cards and visualizations.

Total Sales

Total Sales = SUM(Amazon_Data[Sales])

Total Orders

Total Orders = DISTINCTCOUNT(Amazon_Data[Order_ID])

Total Quantity

Total Quantity = SUM(Amazon_Data[Quantity])

Average Order Value

Avg Order Value = DIVIDE([Total Sales], [Total Orders])

Total Customers

Total Customers = DISTINCTCOUNT(Amazon_Data[Customer_Name])

Sales MoM % Change

Sales MoM % Change =

```
DIVIDE(
    [Total Sales] -
    CALCULATE(
        [Total Sales],
        DATEADD(Amazon_Data[Order_Date], -1, MONTH)
    ),
    [Total Sales],
    1
```

```
CALCULATE(  
    [Total Sales],  
    DATEADD(Amazon_Data[Order_Date], -1, MONTH)  
)  
)
```

Top Category Sales

Top Category Sales =

```
CALCULATE(  
    [Total Sales],  
    TOPN(  
        1,  
        VALUES(Amazon_Data[Category]),  
        [Total Sales]  
    )  
)
```

% of Total Sales

% of Total Sales =

```
DIVIDE(  
    [Total Sales],  
    CALCULATE(  
        [Total Sales],  
        ALL(Amazon_Data)  
    )  
)
```

The DAX measures are calculated dynamically according to the current filter context.

7. Dashboard Visual Design

Page 1 — Sales Overview

Visual	Fields	Purpose
Line Chart	Order Month + Total Sales	Monthly sales trend
Donut Chart	Payment Method + Total Sales	Revenue by payment method
Bar Chart	Region + Total Sales	Regional revenue comparison

Page 2 — Product & Category Analysis

Visual	Fields	Purpose
Horizontal Bar	Category + Total Sales	Category revenue comparison
Column Chart	Top 5 Product + Total Sales	Top products
Clustered Column	Category + Total Quantity	Category-wise unit comparison

Page 3 — Regional & Customer Analysis

Visual	Fields	Purpose
Bar Chart	Region + Total Orders	Orders by region
Donut Chart	Region + Total Sales	Regional sales contribution
Column Chart	Payment Method + Total Orders	Payment usage by order count

The dashboard visual-to-business-question mapping follows the final project presentation.

8. Dashboard KPIs

The final dashboard displays key metrics such as:

- **Total Sales:** ₹6.29 Cr
- **Total Orders:** 5,820
- **Average Order Value:** ₹11,029
- **Units Sold:** 31,344

- **Categories:** 10
- **Products:** 68
- **Unique Customers:** 599
- **Regions:** 5

These values represent the cleaned dashboard dataset shown in the final project presentation.

9. Key Analytical Findings

Product and Category

Furniture is the highest-revenue category, generating approximately **₹2.15 Cr** in sales. However, it has the lowest unit volume among the categories shown, demonstrating that high revenue does not always correspond to high sales volume.

Regional Performance

South records the highest regional sales at approximately **₹1.29 Cr**, while the five regions have relatively close sales values.

Payment Methods

Credit Card and Debit Card generate the highest sales among the payment methods, while UPI also represents a significant share of sales.

Data Quality

The cleaning process removed duplicate records and addressed incomplete records before calculating the final dashboard KPIs.

10. Performance and Usability

The solution is designed to provide:

- Interactive filtering
- Dynamic KPI calculations
- Clear visual comparison
- Consistent calculations across pages
- Easy navigation between dashboard sections
- Reusable DAX measures
- Automated Power Query transformations during refresh

11. Refresh Process

The solution can be refreshed using the Power BI refresh functionality.

The intended refresh workflow is:

Updated Source Data



Power Query Transformations



Data Model Refresh



DAX Recalculation



Updated Dashboard

This reduces the need to manually recreate calculations after updating the source data.

12. Security and Data Considerations

The dashboard is designed for analytical reporting purposes.

Important considerations include:

- Validate source data before refresh.
- Maintain consistent column names and data types.
- Ensure customer information is handled appropriately.
- Review missing and invalid records during data preparation.
- Avoid exposing unnecessary customer-level information in publicly shared reports.

13. Tools and Technologies

Technology	Purpose
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Microsoft Power BI	Dashboard development
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Technology	Purpose
Power Query	Data cleaning and transformation
DAX	Measures and calculations
Excel/CSV	Source data
GitHub	Project documentation and portfolio

14. Final Solution

The completed solution provides a three-page interactive reporting dashboard covering:

Sales Overview → Product & Category Analysis → Regional & Customer Analysis

The project demonstrates the complete Power BI development lifecycle from raw data preparation through data modeling, DAX calculations, visualization, and business insight generation.

The final deliverables include the source dataset, cleaned Power Query model, calculated columns, DAX measures, nine charts across three dashboard pages, and project documentation.